How much can the fingers lift?

Grasp model for tapered, curling pneumatic fingers with frictional contact

Rudranarayan Pradhan · October 2026

1 Question

The actuator should curl, at least over part of its length, into a hook or spiral that can close round an object. The question asked here is: *for a given friction coefficient μ between finger and object, how large an upward force can the curled fingers exert on a rigid object before it slips out?* The answer is read directly from the contact reaction forces that the (rigid) object exerts on the fingers.

2 Model

2.1 Finger

Each finger is a silicone tube with a hoop-thread winding (so it cannot balloon) and an inextensible strain-limiting layer along the side that faces the object. The layer pins the neutral axis, so the finger is an inextensible planar rod whose reference line is the layer, which is also the contact surface. Cut the finger at arc length s and keep the tip piece together with the air inside it: the air beyond the cut pushes with $P\pi r(s)^2$ through the cavity centre, a lever $d(s)$ from the layer. Moment balance gives

$$m(s) = EI(s)\,\big[\kappa(s) - \kappa_0(s)\big], \qquad \kappa_0(s) = \frac{P\,\pi r(s)^2\,d(s)}{EI(s)}, \tag{2.1}$$

where m is the moment of the contact forces beyond s and EI is the bending stiffness about the layer. Pressure therefore acts as a *spontaneous curvature* κ_0, and contact forces bend the finger away from it. For a concentric cavity of radius $r = R - t$ in a tube of outer radius R and wall t, with the layer on the outer surface ($d = R$),

$$I = \tfrac{\pi}{4}(R^4 - r^4) + \pi(R^2 - r^2)R^2, \qquad \kappa_0 \xrightarrow{t \ll R} \frac{P}{3\,E\,t}. \tag{2.2}$$

The wall thickness, not the outer radius, sets how tightly a section curls. Tapering only the outer radius leaves κ_0 almost unchanged and mainly softens the tip. To make the tip curl more tightly than the base, the wall must thin toward the tip as well; alternatively, the strain-limiting layer can be placed only on the distal part, so that only the tip curls.

2.2 Designs compared

All fingers are 100 mm long, $E = 0.15$ MPa (a soft silicone such as Dragon Skin 10), with a hoop winding and a concentric cavity.

Design	Section	Layer
uniform	$R = 10$ mm, $t = 2$ mm	full length
tapered	R: 10 mm to 5 mm, t: 2 mm to 1 mm (scaled copy)	full length
tip-hook	$R = 10$ mm, $t = 2$ mm	last 40 % only

In the tapered finger the section shrinks in proportion, so $\kappa_0 \propto 1/R$ and the tip curls twice as tightly as the base, giving a spiral (Figure 1).

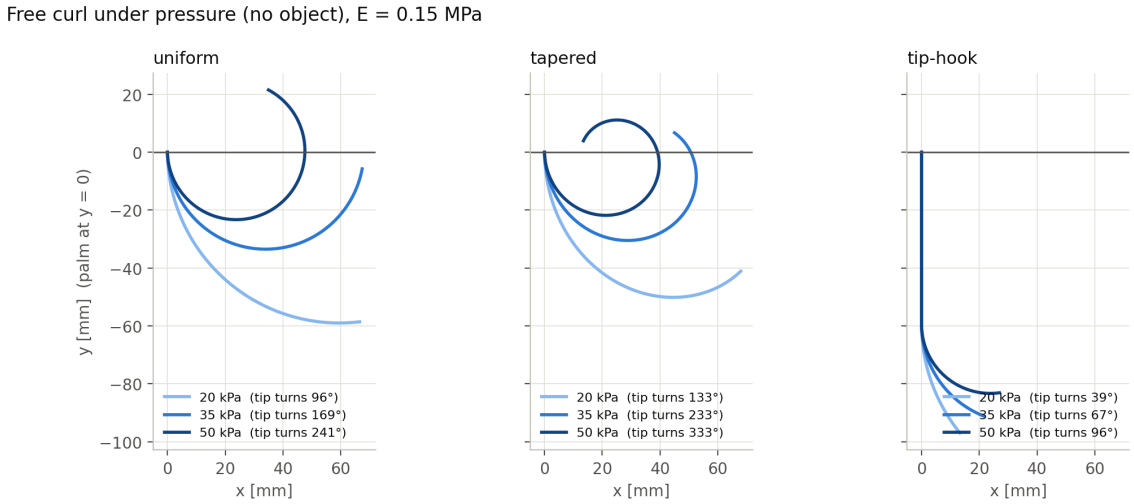


Figure 1: Free curl (no object) at three pressures. Uniform: circular arc. Tapered: tightening spiral. Tip-hook: straight base, curled tip.

2.3 Grasp

Two mirrored fingers hang from a palm at $y = 0$, the layers 1 mm from a rigid cylinder of radius $a = 15$ mm whose centre lies on the axis at depth h below the palm. By symmetry only the left finger is solved; the plane $x = 0$ acts as a frictionless wall where the two fingertips may meet beneath the object.

Contact. Each rod node $\boldsymbol{p}_i$ is pushed out of the cylinder by a penalty force $F_{n,i} = k_n\langle a - |\boldsymbol{p}_i - \boldsymbol{c}|\rangle$. Friction follows Coulomb's law with stick and slip: each contact node carries an anchor ξ_i on the object surface, a tangential spring $k_t(u_i - \xi_i)$ acts while it sticks, and the anchor slides when the spring force would exceed $\mu F_{n,i}$. Within a load step this is the convex, continuously differentiable potential

$$\psi_i = \begin{cases} \tfrac{1}{2} k_t\, \delta_i^2, & |\delta_i| \le \mu F_{n,i}/k_t \\ \mu F_{n,i}\big(|\delta_i| - \tfrac{\mu F_{n,i}}{2k_t}\big), & \text{otherwise,} \end{cases} \qquad \delta_i = u_i - \xi_i, \tag{2.3}$$

with $F_{n,i}$ updated by fixed-point iteration until it changes by less than 10^{-5} N.

Loading. (1) The pressure rises from 0 to P with the object held still, so the fingers close on it. (2) The object is then pulled down in steps (displacement control). At every step the force the fingers exert on the object is obtained from the contact potentials, $\boldsymbol{F} = -\partial(U_n + \Psi)/\partial \boldsymbol{c}$, and split into its normal and friction parts. Doubling the vertical component (two fingers) gives the upward force on the object; its maximum over the pull is the **holding capacity**, i.e. the largest weight the grasp can carry.

Solution. 100 rod segments; for each step the total potential (bending, contact, friction) is minimised with a trust-region Newton method using exact gradients and Hessians from JAX (`grasp_model.py`). One grasp, inflation plus pull, takes about 2 s.

2.4 Checks

- Free curl: the solver reproduces the closed-form integral of κ_0 to machine precision (tip angle 192.64° at 40 kPa, tip position error 10^{-13} mm).
- Force consistency (frictionless): the reaction force summed from the contacts equals the derivative of the stored energy with respect to the object position, computed by finite differences.

- Numerical parameters: doubling the contact stiffness changes the holding force by 0.1 %; refining the rod from 100 to 160 segments changes it by 0.35 %.
- Convergence: every step solves to a gradient below 10^{-9}. The friction–normal-force coupling is iterated with under-relaxation; in 379 of 420 runs every step closes to below 10^{-3} N. The remaining steps all lie in the final slip-out at high friction, well after the holding force is reached (residual at the peak $\leq 3 \times 10^{-3}$ N), and the peaks agree with an independent first pass to within 0.01 N.

3 How the holding force is measured

The object is pulled down from the position where it was grasped. Two numbers are read from the force curve:

Working holding force the largest upward force reached within 10 mm of sag. This is the weight the grasp carries without the object dropping noticeably, and it is the design number.

Ultimate pull-out force the peak over the whole pull, which may need the object to drop by several centimetres first. It relies on the strain-limiting layer being perfectly inextensible and is an upper bound.

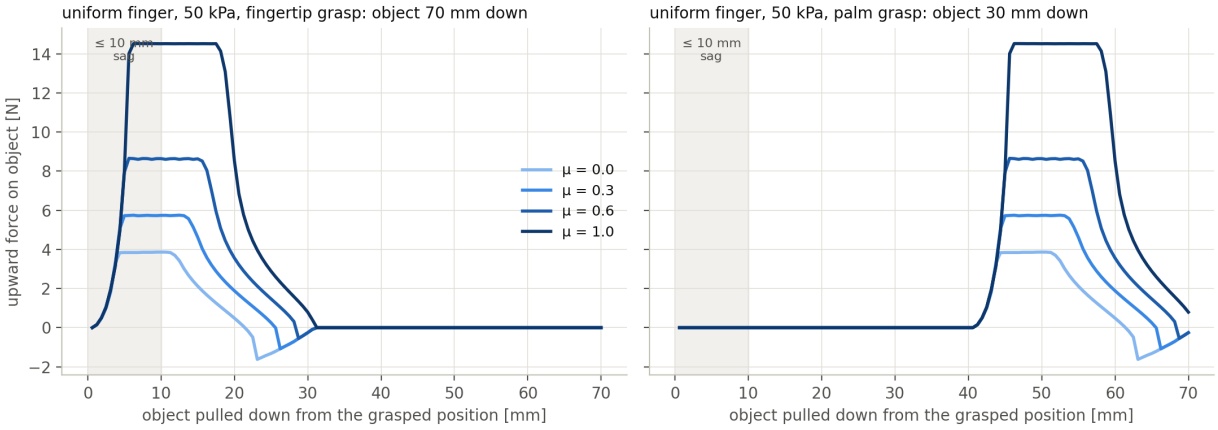


Figure 2: Upward force on the object as it is pulled down, uniform finger at 50 kPa. Left: object near the fingertips; the tips are already under it and the force rises to its plateau within 6 mm. Right: object near the palm; it hangs loose in the closed fingers and drops 40 mm before the tips catch it. The plateau is the hooked fingertips sliding round the object at the friction limit; the shaded band is the 10 mm sag window.

4 Results

The sweep covers the three designs, object depths 20 mm to 80 mm (step 10 mm), pressures 10 kPa to 50 kPa and $\mu = 0$, 0.3, 0.6, 1.0: 420 grasps (`grasp_study.py`).

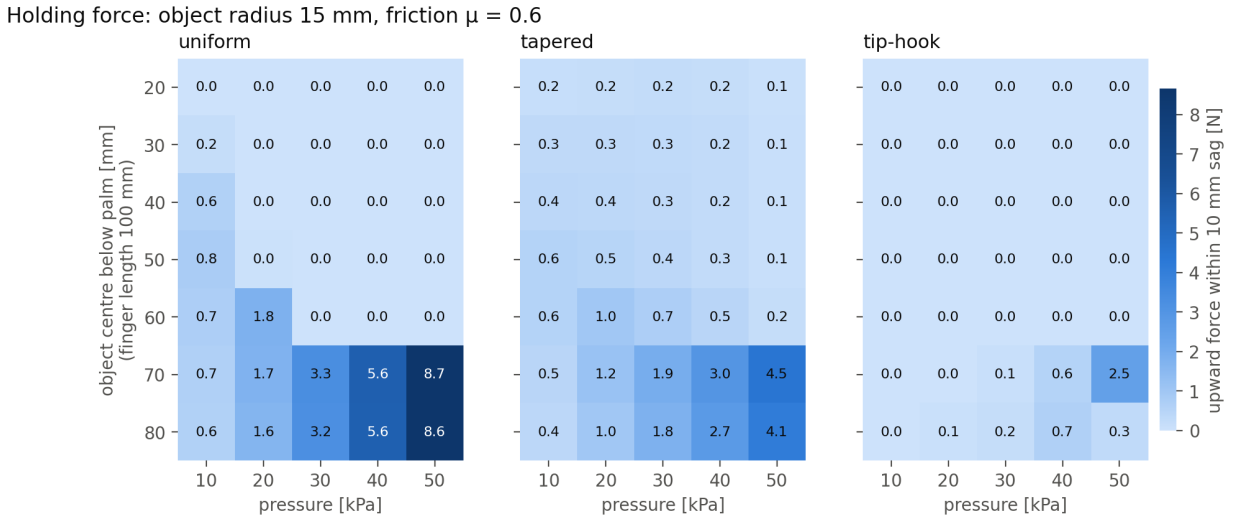


Figure 3: Working holding force against object depth and pressure, $\mu = 0.6$. Only objects near the fingertips are held firmly; objects near the palm are caged loosely and give no force within 10 mm of sag.

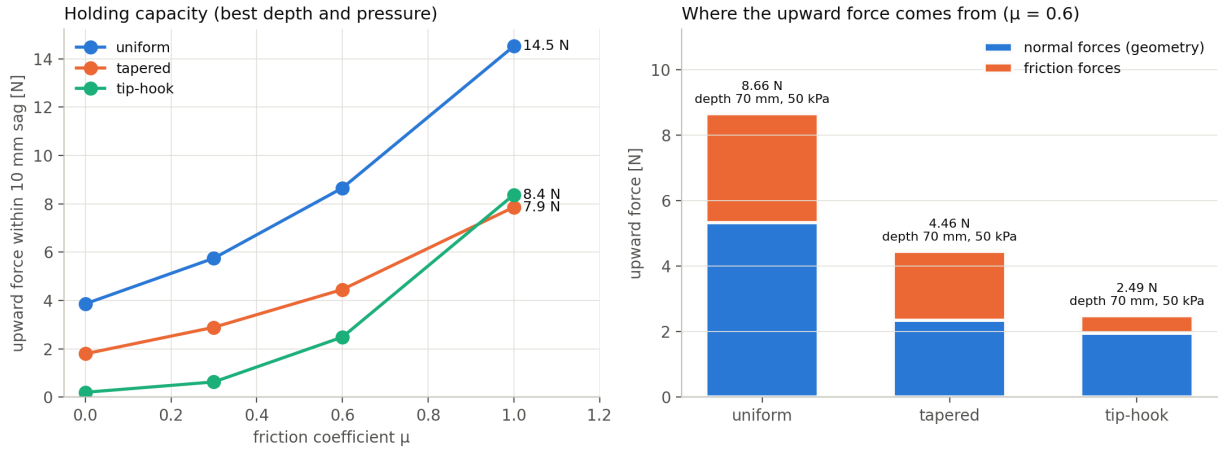


Figure 4: Left: best working holding force against friction. Right: its split into the vertical components of the normal and the friction contact forces at $\mu = 0.6$.

	holding force [N]			
Design	$\mu = 0$	$\mu = 0.3$	$\mu = 0.6$	$\mu = 1.0$
uniform	3.87	5.75	8.66	14.53
tapered	1.80	2.89	4.46	7.86
tip-hook	0.20	0.63	2.49	8.38

Table 1: Best working holding force (two fingers, 50 kPa, object 70 mm to 80 mm below the palm). With the object near the fingertips the ultimate pull-out force equals the working force, because the plateau is reached within 10 mm.

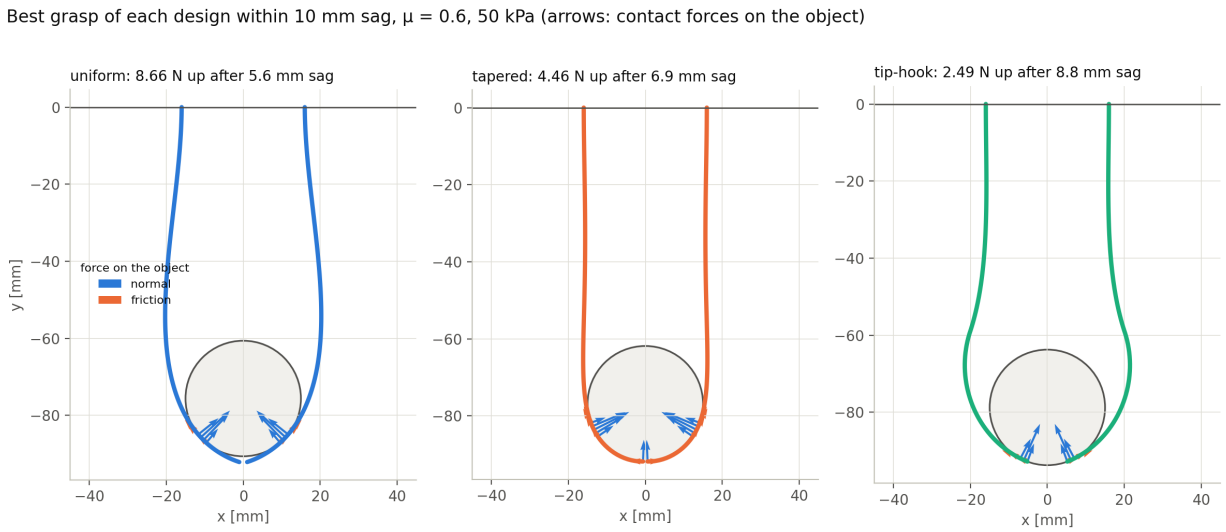


Figure 5: Best grasp of each design at the working holding force, $\mu = 0.6$. Arrows: contact forces on the object. The fingertips curl under the object; nearly all of the upward force acts on its lower third.

1. **Grasp near the fingertips, not near the palm.** The firm grasp is a *fingertip hook*: the object sits where the fingers curl, roughly 70 % to 80 % of the finger length below the palm, and the curled tips are already under it. An object near the palm hangs loose in the closed fingers (they meet beneath it) and drops 40 mm to 50 mm before the tips catch it (Figure 2, right).

2. **Friction roughly doubles to quadruples the grip.** For the uniform finger the working force rises from 3.9 N at $\mu = 0$ to 8.7 N at $\mu = 0.6$ and 14.5 N at $\mu = 1$; at $\mu \geq 0.6$ friction supplies more than a third to over half of the upward force (Figure 4). A tacky or textured contact surface is the cheapest way to lift more.

3. **Pressure helps more than linearly.** At the best depth the uniform finger holds 0.7, 1.7, 3.3, 5.6 and 8.7 N at 10–50 kPa ($\mu = 0.6$): both the curl and the stiffness of the curled tip grow with pressure.

4. **Tapering the outer radius halves the grip.** The tapered finger curls into a tighter spiral but its tip is $2^4 = 16$ times less stiff, so the hook opens under a smaller load: 4.5 N against 8.7 N at $\mu = 0.6$. A taper is useful for wrapping objects of different sizes, not for holding force.

5. **A short tip hook is weak unless friction is high.** With the layer on only the last 40 mm the hook turns about 96° at 50 kPa, too little to get under a 30 mm object; it relies on friction (0.2 N at $\mu = 0$, 8.4 N at $\mu = 1$). A thinner wall in the hook would let it curl further.

5 Segmented fingers

A finger with n sealed chambers in series, each fed separately, has a piecewise-constant spontaneous curvature: a cut inside chamber k sees only P_k, so the finger bends only where its chambers are inflated. The model takes one pressure per chamber (`Finger(n_seg=4)`, `step(P=[P1, P2, P3, P4])`). Inflating single chambers places the bend anywhere along the finger, adjacent chambers make a hook or a spiral, and inflating them in sequence moves the bend from base to tip (animation `report/animations/tentacle_segments`).

Four chambers of 25 mm, wall 1.5 mm, 50 kPa: the free finger curls 348°. With the object near the fingertips this finger holds 14.2 N at $\mu = 0.6$, against 8.7 N for the uniform 2 mm-wall finger: the

thinner wall curls more tightly and hooks further under the object.

Inflation order (`grasp_sequence.py`). Inflating all chambers together, base to tip, or tip to base gives the same grip to within 0.2 % in almost every case, *except* when the object sits at the fingertips: inflating the tip first curls the tip before the finger has closed and it ends up outside the object (0 N). Inflate from the base, or all together. The animation `tentacle_grasp` shows a base-to-tip grasp and the pull-out.

6 Limitations

- Planar model of a cylindrical object with two mirrored fingers; no out-of-plane wrapping, no palm contact, no friction between the fingertips where they meet.
- Linear beam law with a perfectly inextensible strain-limiting layer and a perfect hoop winding (no ballooning). Beyond a curl of about half a turn the strains are large and the 3-D FE model should be used to confirm the numbers.
- Rigid object, point-wise penalty contact on the layer line; finger self-weight is neglected (about 0.1 N per finger, small next to the holding forces).

7 Next steps

1. Measure μ between the cured silicone and typical objects (inclined-plane test) and the silicone's E.
2. Check the best fingertip grasp with the 3-D FE model (add the strain-limiting layer and contact) and in ANSYS.
3. Build one segmented finger; test the holding force by hanging weights from a grasped cylinder and compare with Table 1.

Code: `grasp_model.py` (model), `grasp_study.py` (sweep), `grasp_sequence.py` (inflation order), `grasp_figures.py` (figures), `tentacle_animation.py` (animations).